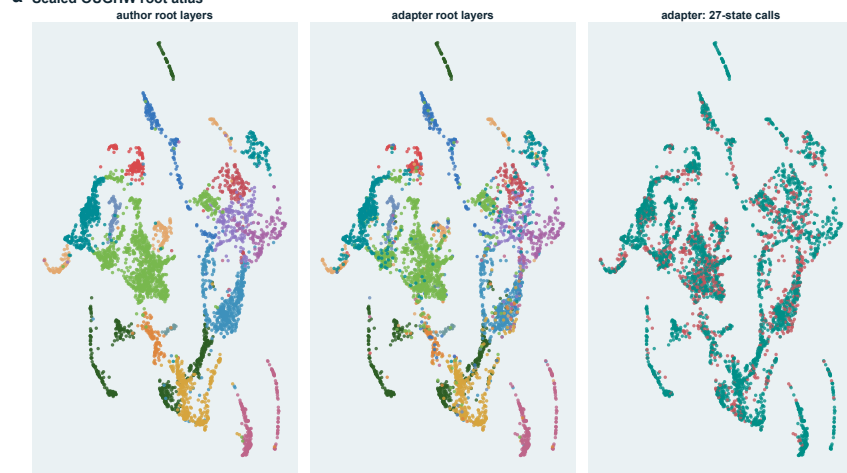


Sorghum root atlas: a physically sealed library recovers 27 cell states

4 independent libraries | OUGHW withheld from fitting | target-sequence LoRA

a Sealed OUGHW root atlas

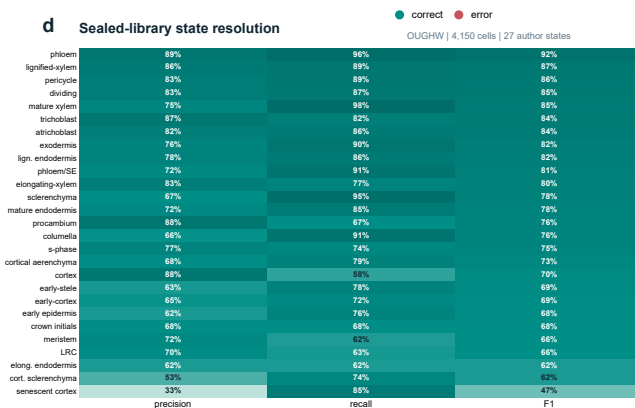


● atrichoblast ● endodermis ● stele
● cortex ● phloem ● xylem

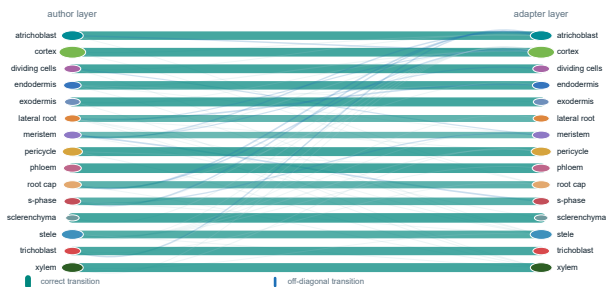
c Orthogroup feature transfer



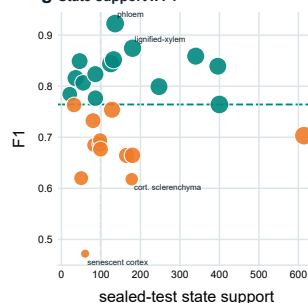
d Sealed-library state resolution



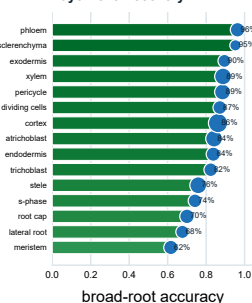
f Root-layer transition network



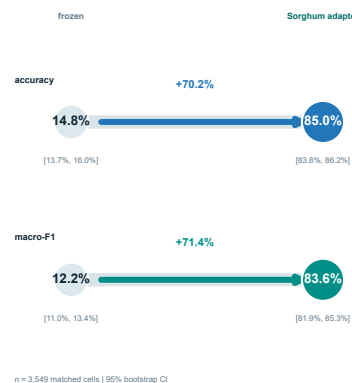
g State support x F1



h Layer-level recovery



b A sealed library recovers both annotation levels



n = 3,540 matched cells | 95% bootstrap CI

e Independent-library contract

